



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: August 03, 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : CE G617
Course Title : Advance Structural Analysis
Instructor-in-charge : SHAMSHER BAHADUR SINGH
[Team of Instructors] : None

1. Course description: [as per the Bulletin]

Flexibility Method; stiffness method; beam curved in plan; two dimensional and three dimensional analysis of structures; shear deformations, shear wall analysis; interactive software development for analysis of structures.

2. Scope and Objective:

Advanced structural analysis techniques and methods provide efficient means and tools for analysis and design of framed structures using matrix approach. Hence learning advance structural analysis will be very useful for structural analyst and designers using computer oriented approach. The basic objective of this course is to acquire the fundamental concepts of matrix methods of structural analysis and their implementation in development of computer programs for computer aided structural analysis of structures. By the end of the course, the students will be familiar with matrix methods of structural analysis using self-developed computer programs wherein both the concepts of flexibility method and stiffness method of structural analysis will be emphasized

3. Course learning outcomes:

- a. Development of softwares for analysis of framed/skeleton structures.
- b. Capability development for the use Matrix based structural analysis principles and formulations for framed structure.

- c. Development of capability to analyze curved members and non-prismatic members using stiffness matrices derived from flexibility matrices.

4. Text books and references:

TB1. William, W. and Gere, J.M., "Matrix Analysis of Framed Structures," CBS Publishers and Distributors, 1986.

RB1. Ghali A. and Neville A.M., "Structural Analysis: A unified classical and Matrix approach,". Chapman and Hall, London, Third Edition. 1989

RB2. Harold C. Martin, "Introduction to Matrix Methods of Structural Analysis," McGraw Hill. Inc. 1966.

RB3. Menon, Devdas, "Advanced Structural Analysis," Narosa Publishing House Pvt. Ltd., Kolkata, 2009.

5. [Method of conduct of the course]

The course will be conducted primarily through Chalkboard as well online medias such as Google class and/or Nalanda.

6. Course Plan: weekly schedule for 14 weeks with [3] lectures per week

Week no.	Module	Lecture no.	Reference	Learning Outcomes
1.	Introduction to Advance Structural Analysis	1-3	[TB, Ch#1]	Definition of the static and kinematic degrees of indeterminacy, General methods of analysis of Indeterminate Structures
2.	Selection of Methods of Structural Analysis and Concept of Equivalent Joint Loads	4-5	[TB, Ch#1]	Understanding of the different methods of structural analysis and significance of evaluating the equivalent joint loads from applied member loads
3.	Fundamentals of Stiffness method	6-8	[TB, Ch#3]	Understanding the concept of basic stiffness method of structural analysis, Effect of Temperature Changes, Prestains, and support displacements on the structural analysis outputs
4.	Formalization of stiffness method and solutions of relevant problems	9-12	[TB, Ch#3]	Application of Basic and formalized stiffness method for indeterminate structural analysis with examples



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5.	Computer Oriented Direct Stiffness method	13-15	[TB, Ch#4]	Learning the concept of direct stiffness method and how it is different from basic stiffness method.
6.	Formulation of member and joint stiffness matrices for beams and Plane Truss structures	16-18	[TB, Ch#4]	Evaluation member and joint stiffness matrices for beams and plane truss structures, and problem solution procedure with emphasis on computer programs.
7.	Formulation of member and joint stiffness matrices for Plane frames	19-21	[TB, Ch#4]	Evaluation of member and joint stiffness matrices for plane frame structures, problem solution procedure with emphasis on computer programs.
8.	Formulation of member and joint stiffness matrices for Grillage structures	22-24	[TB, Ch#4]	Evaluation of member and joint stiffness matrices for Grillage structures, and problem solution procedure with emphasis on computer programs.
9.	Formulation of member and joint stiffness matrices for space truss and frame structures	25-27	[TB, Ch#4]	Evaluation of member and joint stiffness matrices for Space truss and frame structures, and problem solution procedure with emphasis on computer programs.
10.	Fundamentals of Flexibility Methods	28-30	[TB, Ch#2]	Learning the concept of basic flexibility methods and applications for solving beam problems
11.	Use of basic flexibility methods to analyze the truss and plane frame structures	31-33	[TB, Ch#2]	Application of basic flexibility method for solving the problems of truss and plane frame structures

12.	Use of basic flexibility methods to analyze the grillage structures	34-36	[TB, Ch#2]	Application of basic flexibility method for solving the problems of grillage structure
13.	Effect of Temperature changes, prestrains, support displacements and formalized version of flexibility method	37-39	[TB, Ch#2]	Evaluation of the effect of non-mechanical loads on structural response parameters while using flexibility methods and learning the concept of formalized version of flexibility methods and its significance for deriving stiffness matrices and solution of structural analysis problems.
14.	Additional Topics on Stiffness method of structural analysis	40-42	[TB, Ch#6] & [RB3, Ch#7]	Learning stiffness method based analysis concept for curved members, and incorporating the effect of oblique supports, elastic supports, translation of axes, and to utilize the effect of symmetry in the structures and applied loads for analysis.

7. Evaluation Scheme:

Component	Weightage	Duration	Date and Time	Comments
[Class discussions]	[10%]	[In Class]	[2 days per week]	[Will be counted as marks for class interaction.]
[Quiz, 2Nos]	[10%]	[50 minutes]	To be announced in the class	[One quiz before the midterm and other before the comprehensive examination]
[Midterm Examination]	[25%]	[90 minutes]	7 th October, 2026, FN2	[Close Book]
[Take Home]	[10%]	[one per week]	-	[Take home]
[Course Project]	[15%]	[Over the semester]	[Midterm: 29/9/2026 Final Submission: 28/11/2026]	[Open Book]



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				[Based on the computer programs]
[Comprehensive Examination]	[30%]	[3 Hrs.]	8 rd December, 2026, AN	[Open Book]

Criterion for NC: If the student gets less than 20% of the average class marks, he/she will be awarded NC.

- Chamber consultation hours:** Wednesday from 4 to 5 PM or based on the appointment by email.
- Makeup Policy:** Makeup for only midterm and comprehensive examinations will be given provided the prior request is made with genuine reason.

[Shamsher Bahadur Singh]

Instructor-in-charge

[CE G617]

(This form must be filled, ratified by the DCA and submitted to AUGSD of the campus. Items in square brackets are to be filled as per the discretion of the IC after discussions with the team of instructors if any.)